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STERNAL CLOSURE SYSTEMS

Abstract

Sternal closure systems are provided for closing a separated sternum following a surgery that requires accessing a patient's chest cavity. An exemplary sternal closure system may utilize a combination of one or more bone plates, cerclages, and fixation devices.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/563,689, which was filed on Mar. 11, 2024. [0002] The application further claims the benefit of U.S. Provisional Application No. 63/686,908, which was filed on Aug. 26, 2024. [0003] This application further claims the benefit of U.S. Provisional Application No. 63/756,965, which was filed on Feb. 11, 2025. [0004] The disclosures of each of the above applications are incorporated herein by reference in their entirety.

BACKGROUND

[0005] This disclosure relates to the field of surgery, and more particularly to sternal closure systems for performing sternal closure procedures.

[0006] A sternotomy is a surgical procedure performed to gain access to a patient's chest cavity, such as when performing heart surgery, for example. The sternotomy typically requires separating the sternum into two parts. The separated sternum must be rejoined or “closed” after completing the surgery. Known sternal closure techniques involve the use of cerclage wires and/or a combination of bone plates and screws for reapproximating and rejoining the sternum bone sections that were separated during the surgery.

SUMMARY

[0007] This disclosure relates to sternal closure systems for closing a separated sternum following a surgery that requires access to a patient's chest cavity. The sternal closure systems may utilize a combination of one or more bone plates, cerclages, and fixation devices.

[0008] An exemplary sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device deployment assembly including a locking tab, a molly anchor, and a deployment pin. When the locking tab is engaged to the fixation device deployment assembly, the fixation device deployment assembly is configured to insert the molly anchor through the opening of the bone plate. When the locking tab is disengaged from the fixation device deployment assembly, the fixation device deployment assembly is configured to cause the deployment pin to move the molly anchor from a relaxed position to a deployed position.

[0009] An exemplary surgical method may include, inter alia, positioning a bone plate relative to a sternum, advancing a molly anchor of a fixation device deployment assembly through an opening of the bone plate and into the sternum while a locking tab is secured to the fixation device deployment assembly, removing the locking tab from the fixation device deployment assembly, and advancing a deployment pin to cause the molly anchor to transition from a relaxed position to a deployed position relative to the sternum.

[0010] An exemplary method may include, inter alia, advancing a molly anchor of a fixation device deployment assembly through an opening of a bone plate while a locking tab is secured to the fixation device deployment assembly, removing the locking tab from the fixation device deployment assembly, and advancing a deployment pin to cause the molly anchor to transition from a relaxed position to a deployed position relative to the bone plate.

[0011] Another exemplary sternal closure system may include, inter alia, a bone plate including a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface. A ridge protrudes from the bone contacting surface and is configured to align and center the bone plate between a first sternum bone section and a second sternum bone section of a sternum.

[0012] Another exemplary sternal closure system may include, inter alia, a bone plate including a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting

surface, and an outer surface opposite of the bone contacting surface. A first longitudinal slot is provided within the first longitudinal section, and a second longitudinal slot is provided within the second longitudinal section.

[0013] Another exemplary sternal closure system may include, inter alia, a bone plate including a through-passage, a locking ferrule received within the through-passage and including a one-way locking mechanism, and a cerclage including a suture received through the locking ferrule and locked relative to the bone plate by the one-way locking mechanism.

[0014] Another exemplary sternal closure system may include, inter alia, a bone plate including an opening, a screw received within the opening and including a head portion, a threaded portion, and a cross-pin mounted within the head portion, and a suture looped around the cross-pin and configured to secure a tissue relative to the bone plate.

[0015] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a carrier configured to selectively attach to and detach from the bone plate. The carrier may be configured to temporarily hold a fixation device relative to the opening such that, when the carrier is attached to the bone plate, the fixation device vertically overlaps the opening. Further, upon application of force, the fixation device is configured to release from the carrier and at least partially pass through the opening to fix the bone plate relative to a sternum.

[0016] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device configured to fix the bone plate relative to a sternum. The fixation device may be configurable between a relaxed position and a deployed position. Further, the fixation device may include first and second deployable legs on opposite sides of a slit. The first deployable leg may be longer than the second deployable leg.

[0017] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device configured to fix the bone plate relative to a sternum. The fixation device may be configurable between a relaxed position and a deployed position. Further, the fixation device may include a molly anchor and a wedge anchor, and the wedge anchor may include an eyelet coupled to suture.

[0018] Yet another example sternal closure system may include, inter alia, a bone plate including an opening, and a pin inserter including at least one pin. Further, the at least one pin may be configured to be selectively advanced out of the pin inserter. The system may further include a fixation device configured to fix the bone plate relative to a sternum. Further, the fixation device may be configurable between a relaxed position and a deployed position. The fixation device may include a molly anchor, and the molly anchor may be moveable from the relaxed position to the deployed position as the at least one pin comes into contact with first and second deployable legs of the molly anchor.

[0019] The embodiments, examples, and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

[0020] The various features and advantages of this disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 schematically illustrates a sternum of a human thorax. A sternal closure system is utilized to reapproximate and rejoin sternum bone sections that were separated as a result of a

sternotomy.

[0022] FIG. 2 is a top view of an exemplary bone plate of a sternal closure system.

[0023] FIG. 3 is a bottom view of the bone plate of FIG. 2.

[0024] FIG. 4 is an end view of the bone plate of FIG. 2.

[0025] FIG. 5 illustrates another exemplary bone plate of a sternal closure system.

[0026] FIG. 6 illustrates another exemplary bone plate of a sternal closure system.

[0027] FIG. 7 illustrates another exemplary bone plate of a sternal closure system.

[0028] FIG. 8 illustrates another exemplary bone plate of a sternal closure system.

[0029] FIG. 9 illustrates another exemplary bone plate of a sternal closure system.

[0030] FIG. 10 illustrates yet another exemplary bone plate of a sternal closure system.

[0031] FIG. 11 illustrates an exemplary fixation device of a sternal closure system.

[0032] FIG. 12 is a top view of the fixation device of FIG. 11.

[0033] FIG. 13 illustrates an initial position of deployable legs of a molly anchor of the fixation device of FIG. 11.

[0034] FIG. 14 illustrates a deployed position of deployable legs of the molly anchor of the fixation device of FIG. 11.

[0035] FIG. 15 illustrates another exemplary fixation device of a sternal closure system.

[0036] FIG. 16 schematically illustrates fixation of a bone plate to a bone using a plurality of fixation devices configured in the manner shown in FIG. 15.

[0037] FIG. 17 is a top view of another exemplary bone plate of a sternal closure system.

[0038] FIG. 18 is a side view of the bone plate of FIG. 17.

[0039] FIG. 19 illustrates an arrangement of locking barbs of a locking ferrule of the sternal closure system of FIGS. 17-18.

[0040] FIG. 20 illustrates an example assembly configured to facilitate alignment of fixation devices relative to a bone plate.

[0041] FIG. 21 is a perspective view of an exemplary pin inserter.

[0042] FIG. 22 is a cross-sectional view of a portion of an example carrier relative to a portion of an example bone plate, and in particular illustrates an engagement tab of the carrier.

[0043] FIG. 23 is a top view of an exemplary bone plate.

[0044] FIG. 24 is a top view of an exemplary carrier.

[0045] FIG. 25 is a cross-sectional view illustrating an example arrangement of a fixation device relative to the carrier.

[0046] FIG. 26 illustrates another exemplary fixation device in a relaxed position.

[0047] FIG. 27 illustrates the fixation device of FIG. 26 in a deployed position.

[0048] FIG. 28 illustrates another exemplary fixation device in a relaxed position.

[0049] FIG. 29 illustrates the fixation device of FIG. 28 in a deployed position.

[0050] FIG. 30 illustrates an exemplary locking ferrule.

[0051] FIG. 31 is a top view of the locking ferrule of FIG. 30.

[0052] FIG. 32 illustrates an exemplary fixation device deployment assembly configured for deploying fixation devices relative to a bone plate.

[0053] FIG. 33 is an enlarged cross-sectional view of select portions of the fixation device deployment assembly of FIG. 32.

[0054] FIGS. 34 and 35 illustrate an exemplary bone plate holding device.

[0055] FIG. 36 illustrates another exemplary bone plate holding device.

[0056] FIG. 37 illustrates another exemplary bone plate holding device.

[0057] FIG. 38 illustrates yet another exemplary bone plate holding device.

DETAILED DESCRIPTION

[0058] This disclosure relates to sternal closure systems for closing a separated sternum. An exemplary sternal closure system may include one or more cerclages formed by suture, one or more bone plates, and a plurality of fixation devices for fixating the bone plate to bone.

[0059] An exemplary sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device deployment assembly including a locking tab, a molly anchor, and a deployment pin. When the locking tab is engaged to the fixation device deployment assembly, the fixation device deployment assembly is configured to insert the molly anchor through the opening of the bone plate. When the locking tab is disengaged from the fixation device deployment assembly, the fixation device deployment assembly is configured to cause the deployment pin to move the molly anchor from a relaxed position to a deployed position.

[0060] In any further embodiment, the fixation device deployment assembly includes a pin inserter and a pusher having a shaft that is receivable within a through-bore of the pin inserter.

[0061] In any further embodiment, the through-bore extends along a longitudinal axis of the pin inserter and extends through both a handle portion and a shaft portion of the pin inserter.

[0062] In any further embodiment, the shaft portion of the pin inserter is accommodated within a cannulation of the molly anchor at a location between a first head portion of the molly anchor and a second head portion of the deployment pin.

[0063] In any further embodiment, the deployment pin is preassembled to the molly anchor such that the second head portion of the deployment pin is countersunk within the cannulation relative to the first head portion.

[0064] In any further embodiment, the second head portion of the deployment pin abuts against a circumferential ledge located within the cannulation of the molly anchor when the molly anchor is in the relaxed position.

[0065] In any further embodiment, when the locking tab is disengaged from the shaft of the pusher, the deployment pin is movable into engagement with the molly anchor to cause a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.

[0066] In any further embodiment, the shaft of the pusher extends from a proximal end section of the pusher, and further wherein the proximal end section includes first diameter that is greater than a second diameter of the shaft.

[0067] In any further embodiment, the locking tab is configured to engage the shaft of the pusher to prevent the shaft from engaging the deployment pin.

[0068] In any further embodiment, when the locking tab is disengaged from the shaft of the pusher, the deployment pin is movable into engagement with the molly anchor to cause a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.

[0069] In any further embodiment, the deployment pin is movably accommodated within a cannulation of the molly anchor.

[0070] In any further embodiment, the molly anchor includes a head portion and a body portion having a first deployable leg and a second deployable leg that each extend distally from the head portion.

[0071] In any further embodiment, the bone plate includes a body including a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface. A ridge protrudes from the bone contacting surface and is configured to align and center the bone plate relative to a bone.

[0072] In any further embodiment, the bone plate includes a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, an outer surface opposite of the bone contacting surface, a first longitudinal slot provided within the first longitudinal section, and a second longitudinal slot provided within the second longitudinal section.

[0073] In any further embodiment, a holding device including a holding fork or a holding ball is configured to engage the bone plate for holding the bone plate down against a bone.

[0074] An exemplary surgical method may include, inter alia, positioning a bone plate relative to a sternum, advancing a molly anchor of a fixation device deployment assembly through an opening of the bone plate and into the sternum while a locking tab is secured to the fixation device deployment assembly, removing the locking tab from the fixation device deployment assembly, and advancing a deployment pin to cause the molly anchor to transition from a relaxed position to a deployed position relative to the sternum.

[0075] In any further embodiment, advancing the molly anchor includes applying a first force to a pusher of the fixation device deployment assembly with the locking tab secured to a shaft of the pusher.

[0076] In any further embodiment, advancing the deployment pin includes applying a second force to the pusher with the locking tab removed from the shaft of the pusher.

[0077] In any further embodiment, applying the second force advances the deployment pin distally into a cannulation of the molly anchor to release the molly anchor from the fixation device deployment assembly.

[0078] In any further embodiment, advancement of the deployment pin distally into the cannulation causes a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.

[0079] An exemplary method may include, inter alia, advancing a molly anchor of a fixation device deployment assembly through an opening of a bone plate while a locking tab is secured to the fixation device deployment assembly, removing the locking tab from the fixation device deployment assembly, and advancing a deployment pin to cause the molly anchor to transition from a relaxed position to a deployed position relative to the bone plate.

[0080] Another exemplary sternal closure system may include, inter alia, a bone plate including a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface. A ridge protrudes from the bone contacting surface and is configured to align and center the bone plate between a first sternum bone section and a second sternum bone section of a sternum.

[0081] Another exemplary sternal closure system may include, inter alia, a bone plate including a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface. A first longitudinal slot is provided within the first longitudinal section, and a second longitudinal slot is provided within the second longitudinal section.

[0082] Another exemplary sternal closure system may include, inter alia, a bone plate including a through-passage, a locking ferrule received within the through-passage and including a one-way locking mechanism, and a cerclage including a suture received through the locking ferrule and locked relative to the bone plate by the one-way locking mechanism.

[0083] Another exemplary sternal closure system may include, inter alia, a bone plate including an opening, a screw received within the opening and including a head portion, a threaded portion, and a cross-pin mounted within the head portion, and a suture looped around the cross-pin and configured to secure a tissue relative to the bone plate.

[0084] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a carrier configured to selectively attach to and detach from the bone plate. The carrier may be configured to temporarily hold a fixation device relative to the opening such that, when the carrier is attached to the bone plate, the fixation device vertically overlaps the opening. Further, upon application of force, the fixation device is configured to release from the carrier and at least partially pass through the opening to fix the bone plate relative to a sternum.

[0085] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device configured to fix the bone plate relative to a sternum. The fixation

device may be configurable between a relaxed position and a deployed position. Further, the fixation device may include first and second deployable legs on opposite sides of a slit. The first deployable leg may be longer than the second deployable leg.

[0086] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a fixation device configured to fix the bone plate relative to a sternum. The fixation device may be configurable between a relaxed position and a deployed position. Further, the fixation device may include a molly anchor and a wedge anchor, and the wedge anchor may include an eyelet coupled to suture.

[0087] Another example sternal closure system may include, inter alia, a bone plate including an opening, and a pin inserter including at least one pin. Further, the at least one pin may be configured to be selectively advanced out of the pin inserter. The system may further include a fixation device configured to fix the bone plate relative to a sternum. Further, the fixation device may be configurable between a relaxed position and a deployed position. The fixation device may include a molly anchor, and the molly anchor may be moveable from the relaxed position to the deployed position as the at least one pin comes into contact with first and second deployable legs of the molly anchor.

[0088] Yet another example sternal closure system may include, inter alia, a bone plate including an opening, and an assembly including a pin inserter, a pusher, a locking tab, and a fixation device that includes a molly anchor and a deployment pin. When the locking tab is engaged with a shaft of the pusher, the assembly is configured to insert the molly anchor through the opening, and when the locking tab is disengaged from the shaft, the assembly is configured to allow the deployment pin to move the molly anchor from a relaxed position to a deployed position.

[0089] FIG. 1 schematically illustrates a sternum **10** of a human thorax or chest **12**. The sternum **10** provides the anterior central part of the chest **12** and includes a manubrium **14**, a body **16**, and a xiphoid process **18**. A plurality of ribs **20** of the chest **12** are either directly or indirectly attached to the sternum **10** and can articulate relative thereto. Clavicles **22** are supported by and can articulate relative to the manubrium **14** of the sternum **10**.

[0090] As schematically indicated by incision line **25**, a sternotomy has previously been performed on the sternum **10** to access organs and/or other tissues located within a patient's chest cavity, such as during a heart surgery, for example. Although described herein in the context of heart surgery, other surgical procedures could necessitate access to the chest cavity, including but not limited to lung surgeries, for example. As a result of the sternotomy, the sternum **10** is divided along the incision line **25** into two sternum bone sections **10A** and **10B**. The separated sternum bone sections **10A**, **10B** must be reapproximated relative to one another and then rejoined together after completing the heart surgery.

[0091] The separated sternum bone sections **10A** and **10B** may be reapproximated and then rejoined by a sternal closure system **24**. The sternal closure system **24** may include one or more cerclages **26**, one or more bone plates **28**, and a plurality of fixation devices **30**.

[0092] Each cerclage **26** may be established by one or more stands of suture **32** that can be arranged to encircle the sternum **10**. In an embodiment, the sutures **32** are strands of suture tape, such as FiberTape®, for example. FiberTape® is a suture product marketed and sold by Arthrex, Inc. However, other types of suture could be utilized for the sutures **32** within the scope of this disclosure.

[0093] The one or more sutures **32** may be passed through and/or be wrapped around the sternum bone sections **10A** and **10B** to form the cerclage **26**. The sutures **32** may be wrapped around the sternum **10** in various patterns, including figure-8 patterns, for example, to form the cerclage **26**. Moreover, as further detailed below, the sutures **32** can be arranged to interface with the bone plate **28** of the sternal closure system **24**.

[0094] The bone plate **28** may be positioned to straddle the incision line **25** on the sternum **10** such that a first portion of the bone plate **28** interfaces with the sternum bone section **10A** and a second

portion of the bone plate **28** interfaces with the sternum bone section **10B**. The bone plate **28** may include a plurality of openings **34** each configured to receive one of the fixation devices **30**. Each fixation device **30** may be inserted through one of the openings **34** and into either the sternum bone section **10A** or the sternum bone section **10B** for securing the bone plate **28** to the sternum **10**. The fixation devices **30** may include screws, anchors, or a combination of screws and anchors. Together, the cerclage **26** and the fixated bone plate **28** cooperate to rejoin the sternum bone sections **10A** and **10B** together.

[0095] FIGS. **2-4** illustrate an exemplary bone plate **28** that can be utilized as part of the sternal closure system **24** of FIG. **1**. The bone plate **28** may be specifically sized and shaped for use relative to the sternum **10**. The bone plate **28** may be provided in various sizes for accommodating varying patient sizes.

[0096] The bone plate **28** may include a length **L**. In an embodiment, the length **L** may be sized to extend across a majority of the superior-to-inferior length of the sternum **10**. In another embodiment, the length **L** may be sized for fixation on a subsection of the sternum **10**, such as the manubrium **14** or the body **16**, for example.

[0097] The bone plate **28** includes a body **36** that can be made from a polyether ether ketone (“PEEK”) material. The PEEK material may enable the body **36** of the bone plate **28** to be anatomically contoured to a particular patient's anatomy. However, the body **36** could be formed from other biocompatible materials or combination of biocompatible materials.

[0098] The body **36** includes a first longitudinal section **38**, a second longitudinal section **40**, and one or more bridging sections **42** that transversely connect between the first and second longitudinal sections **38**, **40**. When the body plate **28** is received relative to the sternum **10** in the manner shown in FIG. **1**, the first longitudinal section **38** interfaces with the sternum bone section **10A**, and the second longitudinal section **40** interfaces with the sternum bone section **10B**. A cut-out **35** may extend between adjacent bridging sections **42** of the body **36**.

[0099] Each of the first longitudinal section **38** and the second longitudinal section **40** may include a plurality of subsections **39**. In an embodiment, the subsections **39** are curved sections of the first and second longitudinal sections **38**, **40**. However, in other implementations, the subsections **39** could be substantially straight across their respective lengths. The bridging sections **42** could also be either straight or curved along their respective lengths.

[0100] The body **36** of the bone plate **28** may further include a bone contacting surface **44** and an outer surface **46** on an opposite side of the bone plate **28** from the bone contacting surface **44**. The bone contacting surface **44** interfaces with the sternum **10** when the bone plate **28** is positioned thereon.

[0101] The bone contacting surface **44** may include one or more ridges **48** (see FIGS. **3** and **4**) that are configured to be placed along the incision line **25** for aligning and centering the bone plate **28** between the sternum bone sections **10A**, **10B**. One or more of the ridges **48** may extend outwardly from each bridging section **42** of the body **36** and can protrude in a direction away from the outer surface **46**.

[0102] The outer surface **46** may include one or more laser lines **45** that are axially aligned with the ridges **48**. The laser lines **45** provide visual cues for aligning and centering the bone plate **28** to the incision line **25**.

[0103] A first portion of the openings **34** of the bone plate **28** may be provided within the first longitudinal section **38** of the body **36**, and a second portion of the openings **34** of the bone plate **28** may be provided within the second longitudinal section **40** of the body **36**. The openings **34** extend completely through the body **36** and therefore open through both the bone contacting surface **44** and the outer surface **46**. In an embodiment, the openings **34** are equally sized openings that are either aligned or staggered relative to one another. However, other configurations are also contemplated, and thus the total number of openings **34** and their specific arrangement within the body **36** are not intended to limit this disclosure.

[0104] The body **36** of the bone plate **28** shown in FIGS. **2-4** includes an infinity or figure-8-like shape. However, other shapes are contemplated within the scope of this disclosure. For example, the body **36** of the bone plate **28** could include a hexagonal shape (see FIG. **5**), an H-shape (see FIG. **6**), a rectangular shape (see FIG. **7**), a square shape (see FIG. **8**), or any other shape that is suitable to be mounted to the sternum **10** or any subsection thereof.

[0105] FIGS. **9-10** illustrate another exemplary bone plate **28-2**. The bone plate **28-2** may include all the features described above with respect to the bone plate **28** of FIGS. **2-4**. The bone plate **28-2** may additionally include a first elongated slot **50** and a second elongated slot **52** formed through the body **36**. The first elongated slot **50** may be disposed between a first outboard rail **54** of the body **36** and a first lateral edge **56** of the first longitudinal section **38**, and the second elongated slot **52** may be disposed between a second outboard rail **58** of the body **36** and a second lateral edge **60** of the second longitudinal section **40**. In an embodiment, the first elongated slot **50** and the second elongated slot **52** extend across a majority of the length of the body **36** (see FIG. **9**). In another embodiment, the body **36** may include two or more first elongated slots **50** and two or more second elongated slots **52** (see FIG. **10**).

[0106] One or more channels **62** may be formed in the outer surface **46** of the body **36**. The channels **62** may extend from the first elongated slot **50** to the second elongated slot **52** and may each bisect transversely across one of the bridging sections **42**. Each channel **62** may thus establish a countersunk region of the outer surface **46**.

[0107] The first elongated slot **50**, the second elongated slot **52**, and the channels **62** may cooperate to allow the one or more sutures **32** of the cerclage **26** to interface with the bone plate **28**. For example, each suture **32** may be shuttled through the first elongated slot **50**, across one of the channels **62**, and through the second elongated slot **52** while forming the cerclage **26** about the sternum **10**. The sutures **32** may slide within the first and second elongated slots **50**, **52** to allow the bone plate **28** to be adjusted in a superior direction or an inferior direction relative to the sternum **10** prior to final fixation of the bone plate **28**. Further, the sutures **32** may provide sufficient tension to hold the bone plate **28** in place before securely fixating the bone plate **28** to the sternum **10** with the fixation devices **30**. The sutures **32** may be tensioned simultaneously with inserting the fixation devices **30** to achieve definitive fixation of the bone plate **28** to the sternum **10**.

[0108] FIGS. **11-14** illustrate features associated with an exemplary fixation device **30** that can be utilized as part of the sternal closure system **24** of FIG. **1** for securing the bone plate **28** down on the sternum **10**. In an embodiment, the fixation device **30** is made from the same type of material as the bone plate **28** (e.g., a PEEK material). However, the fixation device **30** could alternatively be constructed out of a different material than the bone plate **28**.

[0109] The fixation device **30** may include a molly anchor **64** that extends along a longitudinal axis A between a head portion **66**, located proximally, and a body portion **68**, located distally. The head portion **66** may be configured to mate with one of the openings **34** of the bone plate **28** when the molly anchor **64** is inserted therethrough. A cannulation **70** may extend through the head portion **66** and may circumscribe the longitudinal axis A.

[0110] The body portion **68** of the molly anchor **64** may include a first deployable leg **72** and a second deployable leg **74**. The first deployable leg **72** and the second deployable leg **74** may be separated from one another by a slit **76** that is formed in the body portion **68**. The slit **76** may extend from the head portion **66** to a distal tip **78** of the body portion **68**. The cannulation **70** of the head portion **66** may open into the slit **76**.

[0111] The first deployable leg **72** and the second deployable leg **74** may each include a plurality of bone engagement features **80** for facilitating bone fixation once the molly anchor **64** is inserted through the bone plate **28** and into the sternum **10**. In an embodiment, the bone engagement features **80** include barbs or ledges. In another embodiment, the bone engagement features **80** include screw-like threads. However, other configurations are also contemplated within the scope of this disclosure.

[0112] Referring now primarily to FIGS. 13-14, the fixation device **30** may further include a deployment pin **82** that is configured for moving the first and second deployable legs **72, 74** of the molly anchor **64** between a relaxed position **P1** (see FIG. 13) and a deployed position **P2** (see FIG. 14). The molly anchor **64** includes a greater footprint when in the deployed position **P2** compared to the relaxed position **P1**. In an embodiment, the deployment pin **82** is made of a PEEK material.

[0113] During implantation, the deployment pin **82** may be inserted through the cannulation **70** of the head portion **66**. Upon exiting the cannulation **70**, the deployment pin **82** may enter the slit **76** and then engage an inner wall **84** of each of the first and second deployable legs **72, 74**. As the deployment pin **82** is moved further distally within the slit **76**, the deployment pin **82** forces the first and second deployable legs **72, 74** to splay apart from one another in a radially outward direction, thereby moving the legs from the relaxed position **P1** to the deployed position **P2**. In the deployed position **P2**, the first and second deployable legs **72, 74** are positioned to engage the bone and prevent the fixation device **30** from backing out of the opening **34** of the bone plate **28**.

[0114] FIGS. 15-16 illustrate another exemplary fixation device **130** that can be utilized as part of the sternal closure system **24** of FIG. 1 for securing the bone plate **28** down on the sternum **10**. The fixation device **130** may include a screw **86** that extends along a longitudinal axis **A** between a head portion **88** and a threaded body **90**. The head portion **88** may be configured to mate with one of the openings **34** of the bone plate **28**, and the threaded body **90** may be screwed into underlying bone when the screw **86** is inserted therethrough.

[0115] A cross-pin **92** may be mounted within the head portion **88** of the screw **86**. A suture **94** of the fixation device **130** may be looped around the cross-pin **92**. A needle **96** may be attached to free ends of the suture **94**. The needle **96** may be utilized to pierce through various tissues (e.g., sternopericardial ligament, sternocleidomastoid muscle, etc.) associated with the sternum **10**, and the suture **94** may then be used for securing the tissue relative to the bone plate **28** during the sternal closure.

[0116] FIGS. 17-18 illustrate another exemplary bone plate **128** that can be utilized as part of the sternal closure system **24** of FIG. 1. The bone plate **128** includes a body **136** that includes a first longitudinal section **138**, a second longitudinal section **140**, and one or more bridging sections **142** that transversely connect between the first and second longitudinal sections **138, 140**. The body **136** further includes a bone contacting surface **144** and an outer surface **146** on an opposite side of the bone plate **128** from the bone contacting surface **144**.

[0117] Through-passages **98** may be formed through one or both of the first longitudinal section **138** and the second longitudinal section **140**. Each through-passage **98** may be sized to receive a locking ferrule **99**. The locking ferrules **99** may each be configured for tensioning and knotlessly locking one or more sutures **32** when forming the cerclage **26** around the sternum **10**. Other implementations could involve positioning the through-passages **98** and the locking ferrules **99** within the bridging sections **142** of the body **136**, for example.

[0118] As best illustrated in FIG. 19, each locking ferrule **99** may include a cannulation **100** that establishes an internal passageway for accommodating one or more strands of the suture **32**. A plurality of locking barbs **102** may protrude inwardly from an inner diameter wall **104** that circumscribes the cannulation **100**. The locking barbs **102** may therefore occupy at least a portion of the open space of the cannulation **100**.

[0119] The locking barbs **102** may be arranged in multiple rows along the length of the cannulation **100**. For example, the locking barbs **102** may be arranged in at least a first row **R1** and a second row **R2**. In an embodiment, the locking barbs **102** of the second row **R2** are staggered relative to the locking barbs **102** of the first row **R1**.

[0120] Each locking barb **102** may include a sharp or pointed tip **106**, and each locking barb **102** may be angled in a common direction. The locking barbs **102** may therefore establish a one-way locking mechanism that permits the suture(s) **32** to pass through the cannulation **100** in a first direction **D1** while preventing the suture(s) **32** from being tensioned or otherwise moved in a

second direction D2. The staggered relationship of the rows of locking barbs **102** may provide for maximum engagement with the suture **32** one it is passed through the cannulation **100**.

[0121] During the sternal closure, the suture(s) **32** may be wrapped around the sternum **10** and then passed through the cannulation **100** of the locking ferrule **99**. The suture(s) **32** may then be tensioned to lock the suture(s) **32** in place and thereby form the cerclage **26** around the sternum **10**.

[0122] FIG. **20** illustrates an assembly **150** configured to facilitate deployment of fixation devices **30** relative to the bone plate **28**. Parts of the assembly **150** can be used as part of the sternal closure system **24** of FIG. **1** for securing the bone plate **28** relative to the sternum **10**. While bone plate **28** and fixation devices **30** are shown in FIG. **20**, the assembly **150** can be used with the bone plate **128**, fixation devices **130**, and other bone plates or fixation devices.

[0123] In this example, the assembly **150** includes a pin inserter **152**, a pusher **154**, a locking tab **156**, and a carrier **158**. The fixation devices **30** and the bone plate **28** may be considered part of the assembly **150**.

[0124] The pin inserter **152** is shown in greater detail in FIG. **21**, which illustrates the pin inserter **152** in cross-section. The pin inserter **152** extends along an axis Z between a proximal end **160** and a distal end **162**. The pin inserter **152** may include a handle portion **164** adjacent the proximal end **160**, and may further include a shaft portion **166** projecting from the handle portion **164** and extending from the handle portion **164** to the distal end **162**. The shaft portion **166** exhibits a lesser diameter than the handle portion **164**. The handle portion **164** may be sized and shaped so as to be comfortably gripped by a user. The pin inserter **152** may be fully cannulated, including a through-bore **168** extending from the proximal end **160** to the distal end **162**. In this example, the through-bore **168** is loaded with a plurality of deployment pins **82**. While a particular number of deployment pins **82** are shown in FIG. **21**, this disclosure is not limited to a particular number of deployment pins **82**.

[0125] With reference back to FIG. **20**, the pusher **154** includes a proximal end section **170** and a shaft **172** projecting distally from the proximal end section **170**. The proximal end section **170** exhibits a greater diameter than the shaft **172**. The shaft **172** may be sized and shaped to enter into the through-bore **168** and to engage a proximal-most one of the deployment pins **82** to advance them distally within the through-bore **168** and to eject the distal-most one of the deployment pins **82** from the through-bore **168**. The proximal end section **170** is sized and shaped so as to facilitate and endure a transfer of force from another object, such as a mallet. The locking tab **156** may be configured to selectively engage and disengage a portion of the shaft **172** projecting distally of the pin inserter **152** to prevent unwanted advancement of the pusher **154**.

[0126] The carrier **158** may be configured to support the fixation devices **30** relative to the bone plate **28** so as to facilitate alignment of the fixation devices **30** relative to the bone plate **28** and facilitate advancement of the fixation devices **30** into the sternum, in this example.

[0127] The carrier **158**, in this example, includes a main housing section **174** including a top surface **176** and a bottom surface **178** spaced-apart from one another by a height H. When mounted to a bone plate **28** arranged relative to sternum **10**, the top surface **176** may be referred to as an anterior surface, and the bottom surface **178** may be referred to as a posterior surface. The height H may be substantially equal to a length dimension of the fixation devices **30**. The carrier **158** further includes engagement tabs **180**, which in this example project in a downward direction relative to the bottom surface **178**. The engagement tabs **180** are configured to snap around an outer edge of the bone plate **28** to selectively attach the carrier **158** to the bone plate **28**, as generally shown in FIG. **22**. The engagement tabs **180** may be arranged about the perimeter of the carrier **158** so as to facilitate alignment of the carrier **158** relative to the bone plate **28**. While engagement tabs **180** are shown and described, this disclosure extends to other techniques for aligning the carrier **158** relative to the bone plate **28**.

[0128] As seen in FIGS. **23** and **24**, the carrier **158** exhibits a shape substantially similar to that of the bone plate **28**. While a particular configuration of the bone plate **28**, and a corresponding

configuration of the carrier **158**, is shown in FIGS. **23** and **24**, this disclosure extends to other bone plate configurations, and other corresponding carriers, which would be sized and shaped to correspond to the particular bone plate.

[0129] The bone plate **28** includes six openings **34**, in this example. When the carrier **158** is aligned such that the carrier **158** vertically overlaps the bone plate **28**, namely when the tabs **180** are engaged with the bone plate **28**, the fixation devices **30** are arranged above a corresponding one of the openings **34**. Namely, axes A of the fixation devices **30** are substantially concentric with a centerline C of a corresponding opening **34**. As shown in FIG. **24**, there are six fixation devices **30**, in this example, with each one corresponding to one of the openings **34**.

[0130] With reference to FIG. **25**, an example configuration of the carrier **158** relative to one of the fixation devices **30** is shown. The carrier **158** may be configured similarly relative to the other fixation devices **30**. In this example, the carrier **158** may be configured to selectively hold the fixation device **30** such that, upon application of force, the fixation devices **30** will break free of the carrier **158**. In this example, the carrier **158** includes a through-bore **182** extending from the top surface **176** to the bottom surface **178**. The through-bore **182** may be sized to as to permit the fixation device **30** to pass through the through-bore **182**. That said, the through-bore **182** is, in this example, in relatively close proximity to the fixation device **30** to prevent premature movement of the first and second deployable legs **72**, **74** to the deployed position. In this example, the fixation device **30** includes the molly anchor **64** of FIGS. **13** and **14**. The carrier **158**, in this example, includes first and second frangible tabs **184**, **186** projecting from the main housing section **174** to the head portion **66** of the molly anchor **64**. The first and second frangible tabs **184**, **186** are configured to release the head portion **66** of the molly anchor **64** upon application of a sufficient force. While frangible tabs are shown and described, this disclosure encompasses other techniques for temporarily holding the fixation devices **30** by the carrier **158**. Further, while the fixation devices **30** are not shown with threads, this disclosure extends to threaded fixation devices. The fixation devices **30** may alternatively include barbs or ledges, as examples.

[0131] In an example method of use, a bone plate **28** may be arranged relative to the sternum **10**. The carrier **158**, with the fixation devices **30** attached thereto, may be attached to the bone plate **28** using the tabs **180**, for example. A user may situate the pin inserter **152** relative to a particular one of the fixation devices **30**, remove the locking tab **156**, and apply a force to the proximal end section **170** of the pusher **154**, such as by using a mallet. Doing so advances one of the deployment pins **82** distally out of the pin inserter **152**, releases the fixation device **30** from the carrier **158** by fracturing tabs **184**, **186**, advances the fixation device **30** partially through a corresponding one of the openings **34** of the bone plate **28**, and causes the first and second deployable legs **72**, **74** to move into the deployed position, as shown in FIG. **14**. The process is repeated for the remaining fixation devices **30**. The carrier **158** may then be detached from the bone plate **28**.

[0132] FIGS. **26** and **27** illustrate another example fixation device **30**, which may be used with the sternal closure system **24**, in the relaxed and deployed positions respectively. In this example, the example fixation device **30** may be a molly anchor **188** and may be configured substantially similar to the molly anchor **64**, with the exception that the molly anchor **188** includes first and second deployable legs **190**, **192** of different lengths. The first deployable leg **190**, which exhibits a greater length than the second deployable leg **192**, may facilitate penetration of the molly anchor **188** into bone. In this regard, the molly anchor **188** may be considered self-punching. The second deployable leg **192** may enter into a bone hole formed by the first deployable leg **190**. The mismatched length of the first and second deployable legs **190**, **192** may provide self-punching functionality while avoiding premature movement of the molly anchor **188** to the deployed position.

[0133] FIGS. **28** and **29** illustrate another example fixation device **30**, which may be used with the sternal closure system **24**, in the relaxed and deployed positions respectively. In this example, the fixation device **30** includes the molly anchor **64**, or an anchor configured substantially similar

thereto, and further includes a wedge anchor **194**. The wedge anchor **194** may be part of the molly anchor **64** or could be incorporated as part of an inserter device (not shown for simplicity and clarity) usable to implant the molly anchor **64**. The molly anchor **64** may be threaded in this example. The fixation device **30** of FIGS. **28** and **29** may be considered self-punching. When the molly anchor **64** is in the relaxed position, the wedge anchor **194** may be arranged adjacent a distal end of the first and second deployable legs **72**, **74**. The carrier **158** may temporarily hold the wedge anchor **194** in this configuration. Alternatively or additionally, the first and second deployable legs **72**, **74** may include one or more features, such as notches or grooves, configured to temporarily hold the wedge anchor **194** in the configuration of FIG. **28**.

[0134] The wedge anchor **194** includes a substantially sharp distal tip **196** to facilitate penetration into bone. The wedge anchor **194** further includes a wedge section **198** tapering toward an eyelet **200** adjacent a proximal end **202** of the wedge anchor **194**. In this example, suture **204** may be threaded through the eyelet **200** and passes through the molly anchor **64** and projects proximally of the head portion **66** of the molly anchor **64** such that the suture **204** can be manipulated by a user. In an example method of use, a force can be applied to the fixation device **30**, thereby causing the distal tip **196** to penetrate bone and to cause the first and second deployable legs to advance into a bone hole formed by the distal tip **196**. A user can then pull the suture **204**, which causes the wedge section **198** to force the first and second deployable legs **72**, **74** into the deployed position. With the fixation device **30** in the deployed position of FIG. **29**, a user may tie soft tissue and/or skin relative to the bone plate **28** via suture **204**. Alternatively or additionally, suture **204** may establish one or more of the cerclages **26**. The suture **204** may be made of the same material as suture **32**.

[0135] FIGS. **30** and **31** illustrate an exemplary locking ferrule **210** that can be used when performing various tensionable surgical procedures, including using the fixation device of FIGS. **28** and **29**, or similar fixation devices. The locking ferrule **210** may be used with the sternal closure system **24**. In particular, the locking ferrule **210** may be provided adjacent to or within through passages in a bone plate, similar to the locking ferrule **99** described above.

[0136] The locking ferrule **210** may be an exemplary suture locking device that can include a cap portion **212** and a barrel portion **214** that protrudes from the cap portion **212**. The cap portion **212** and the barrel portion **214** may be formed together as part of a unitary structure of the locking ferrule **210**. The cap portion **212** may rest against outer surface **46** of the bone plate **28**, and the barrel portion **214** may project into an opening **34** of the bone plate **28**, in an example. In an embodiment, the cap portion **212** and the barrel portion **214** are arranged to provide a top hat-like appearance of the locking ferrule **210**. However, the specific size, shape, and material make-up of the locking ferrule **210** are not intended to limit this disclosure.

[0137] The cap portion **212** of the locking ferrule **210** may include a top surface **216**, a bottom surface **218**, and a side wall **220** that extends between the top surface **216** and the bottom surface **218**. The barrel portion **214** may protrude outwardly from the bottom surface **218** of the cap portion **212**. A cannulation **222** may extend through the cap portion **212** and the barrel portion **214**. The cannulation **222** may establish an internal passageway for accommodating one or more strands of suture or other structures (e.g., wires, shafts, etc.). The locking ferrule **210** of this embodiment may therefore provide a “single-chamber” design.

[0138] A plurality of locking barbs **230** may occupy at least a portion of the open space of the cannulation **222**. In an embodiment, the locking barbs **230** are integrally formed (e.g., molded) features of the body of the locking ferrule **210**. The locking barbs **230** may be provided along an entire length of the cannulation **222** or at only select portions thereof. The locking barbs **230** may be either rigid or flexible structures.

[0139] Each locking barb **230** may include a sharp or pointed tip **232**, and, in this implementation, each locking barb **230** may be angled in a direction toward the cap portion **212**. The locking barbs **230** may therefore establish a one-way locking mechanism that permits one or more sutures to pass through the cannulation **222** in a first direction while preventing the suture(s) from being tensioned

or otherwise moved in a second, opposite direction.

[0140] FIGS. 32-33 illustrate another exemplary assembly 350 configured to facilitate deployment of a fixation device 30 relative to the bone plate 28. The assembly 350 may be referred to as a fixation device deployment assembly. In this embodiment, the fixation device 30 may include the molly anchor 64 and the deployment pin 82, and the assembly 350 can be used as part of the sternal closure system 24 of FIG. 1 for securing the bone plate 28 relative to the sternum 10. While the bone plate 28 and the molly anchor 64 are specifically shown in FIG. 32, the assembly 350 can be used with the bone plate 128 and/or other bone plates and/or other fixation devices.

[0141] The assembly 350 may include a pin inserter 352, a pusher 354, and a locking tab 356. The fixation device 30, which in this embodiment includes both the molly anchor 64 and the deployment pin 82, may also be considered part of the assembly 350. The bone plate 28 may be utilized with the assembly 350 as part of the sternal closure system 24.

[0142] The pin inserter 352 may extend along a longitudinal axis between a proximal end 360 and a distal end 362. The pin inserter 352 may include a handle portion 364 adjacent the proximal end 360 and a shaft portion 366 projecting outwardly from the handle portion 364 to the distal end 362. The shaft portion 366 may exhibit a lesser diameter than the handle portion 364. The handle portion 364 may be sized and shaped so as to be comfortably gripped by a user. The pin inserter 352 may be fully cannulated and may thus include a through-bore 368 extending from the proximal end 360 to the distal end 362.

[0143] The fixation device 30 may be pre-loaded onto the distal end 362 of the pin inserter 352. When positioned thereon, a portion 399 (see FIG. 33) of the shaft portion 366 that provides the distal end 362 of the pin inserter 352 may be accommodated within the cannulation 70 of the molly anchor 64 at a location between the head portion 66 of the molly anchor 64 and a head portion 83 of the deployment pin 82. The head portion 66 of the molly anchor 64 may circumscribe an outer diameter surface of the portion 399 of the shaft portion 366 when the fixation device 30 is preassembled to the pin inserter 352. The deployment pin 82 may be preassembled to the molly anchor 64 such that the head portion 83 of the deployment pin 82 is slightly countersunk relative to the head portion 66 of the molly anchor 64. A circumferential ledge 71 (see FIG. 33) located within the cannulation 70 may prevent the shaft portion 366 of the pin inserter 352 and the head portion 83 of the deployment pin 82 from being inserted too far into the cannulation 70 prior to full deployment of the fixation device 30. For example, the head portion 83 of the deployment pin 82 can abut against the circumferential ledge 71 to prevent advancement of the deployment pin 82 further distally into the cannulation 70 prior to full deployment of the fixation device 30.

[0144] In the illustrated embodiment, the assembly 350 is configured to deploy a one fixation device 30 at a time. However, other configurations are contemplated within the scope of this disclosure.

[0145] The pusher 354 of the assembly 350 may include a proximal end section 370 and a shaft 372 projecting distally from the proximal end section 370. The proximal end section 370 exhibits a greater diameter compared to the shaft 372. The shaft 372 may be sized and shaped for insertion into the through-bore 368 to engage the head portion 83 of the deployment pin 82 for causing the first and second deployable legs 72, 74 of the molly anchor 64 to splay apart from one another after the molly anchor 64 has been inserted into one of the openings 34 of the bone plate 28. The proximal end section 370 may be sized and shaped so as to facilitate and endure a transfer of force from another object, such as a mallet, for example, for forcing the deployment pin 82 further distally into the cannulation 70.

[0146] The locking tab 356 may be configured to selectively engage and disengage the shaft 372 of the pusher 354 to prevent unwanted advancement of the pusher 354. When engaged to the shaft 372, the locking tab 356 may be positioned between the proximal end 360 of the pin inserter 352 and the proximal end section 370 of the pusher 354. The shaft 372 cannot engage the head portion 83 of the deployment pin 82 without first disengaging the locking tab 356 from the shaft 372. Once

the locking tab **356** is removed from the shaft **372**, the shaft **372** can be moved further distally within the through-bore **368** for engaging the head portion **83** of the deployment pin **82**.

[0147] In an example method of use of the assembly **350**, the bone plate **28** may be arranged relative to the sternum **10**. The pin inserter **352**, with the pusher **354**, the locking tab **356**, and the fixation device **30** attached thereto, may be positioned relative to one of the openings **34** of the bone plate **28**. A force **F** may then be applied (e.g., with a mallet) to the proximal end section **370** of the pusher **354** with the locking tab **356** secured to the shaft **372** to advance the molly anchor **64** through the opening **34** of the bone plate **28** and into the sternum **10**. A pilot hole may be formed (e.g., punched, drilled, etc.) in the sternum **10** prior to inserting the molly anchor **64** through the opening **34**.

[0148] A user may next remove the locking tab **356** from the shaft **372** (e.g., by pulling in the direction of arrow **D1**) and then apply an additional force **F2** to the proximal end section **370** of the pusher **354**, such as by using the mallet. Doing so advances the deployment pin **82** further distally into the cannulation **70**, thereby causing the molly anchor **64** to be released from the pin inserter **352** and the first and second deployable legs **72**, **74** to move to the deployed position. In the deployed position, the first and second deployable legs **72**, **74** can engage the sternum **10** and prevent the molly anchor **64** from backing out of the opening **34** of the bone plate **28**. This process may be repeated for inserting and deploying additional fixation devices **30** through the bone plate **28** and into the sternum **10**.

[0149] Although not specifically shown, a carrier, such as the carrier **158**, for example, could be utilized to facilitate alignment of the fixation devices **30** relative to the bone plate **28** prior to insertion of the fixation devices **30** through the bone plate **28**. The carrier **158** may also be used to guide the formation of pilot holes that can be drilled or punched into the sternum **10** prior to inserting the molly anchor **64** through the opening **34**.

[0150] A holding device **400** may be utilized to hold the bone plate **28** in place while inserting the fixation devices **30** in the manner described above. For example, as shown in FIGS. **34** and **35**, the holding device **400** may include a handle **402**, a shaft **404** projecting from the handle **402**, and a holding fork **406** provided at a distal end of the shaft **404**. The holding fork **406** may be sized to be arranged about one of the bridging sections **42** of the bone plate **28** for more firmly holding the bone plate **28** down against the sternum **10** when inserting the fixation devices **30**.

[0151] In another embodiment, shown in FIG. **36**, the holding device **400** may include a holding ball **410** instead of the holding fork **406**. The holding ball **410** may be sized to be positioned within one of the openings **34** of the bone plate **28** for more firmly holding the bone plate **28** down against the sternum **10** when inserting the fixation devices **30**.

[0152] FIG. **37** illustrates another exemplary holding device **500**. The holding device **500** is similar to that of FIGS. **34-35** and thus includes a handle **502**, a shaft **504** projecting from the handle **502**, and a holding fork **506** provided at a distal end of the shaft **504**. However, in this embodiment, the holding fork **506** may be sized to be arranged about one of the longitudinal sections of the bone plate **28** (as opposed to the bridging sections) for more firmly holding the bone plate **28** down against the sternum **10** when inserting the fixation devices **30**.

[0153] FIG. **38** illustrates yet another exemplary holding device **600**. The holding device **600** is similar to that of FIG. **36** and thus includes a handle **602**, a shaft **604** projecting from the handle **602**, and a holding ball **610** provided at a distal end of the shaft **604**. The holding ball **610** may be sized to be positioned within one of the openings **34** of the bone plate **28**. The holding ball **610** may include a spike **612** for engaging and gripping underlying bone for more firmly holding the bone plate **28** down against the sternum **10** when inserting the fixation devices **30**.

[0154] The sternal closure systems of this disclosure may achieve definitive sternal fixation in a manner in which both a cerclage and bone plate can be simultaneously tensioned/implanted. Various components (e.g., plate, anchor, carrier, etc.) of the system may be made of a PEEK material that can be bent to match a patient's anatomy. The bone plates and carriers of the system

may also be anatomically shaped.

[0155] Although the different non-limiting embodiments are illustrated as having specific components or steps, the embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.

[0156] It should be understood that like reference numerals identify corresponding or similar elements throughout the several drawings. It should further be understood that although a particular component arrangement is disclosed and illustrated in these exemplary embodiments, other arrangements could also benefit from the teachings of this disclosure. In addition, the various figures accompanying this disclosure are not necessarily to scale, and some features may be exaggerated or minimized to show certain details of a particular component or arrangement.

[0157] The foregoing description shall be interpreted as illustrative and not in any limiting sense. A worker of ordinary skill in the art would understand that certain modifications could come within the scope of this disclosure. For these reasons, the following claims should be studied to determine the true scope and content of this disclosure.

Claims

1. A sternal closure system, comprising: a bone plate including an opening; and a fixation device deployment assembly including a locking tab, a molly anchor, and a deployment pin, wherein when the locking tab is engaged to the fixation device deployment assembly, the fixation device deployment assembly is configured to insert the molly anchor through the opening of the bone plate, and when the locking tab is disengaged from the fixation device deployment assembly, the fixation device deployment assembly is configured to cause the deployment pin to move the molly anchor from a relaxed position to a deployed position.
2. The sternal closure system as recited in claim 1, wherein the fixation device deployment assembly includes a pin inserter and a pusher having a shaft that is receivable within a through-bore of the pin inserter.
3. The sternal closure system as recited in claim 2, wherein the through-bore extends along a longitudinal axis of the pin inserter and extends through both a handle portion and a shaft portion of the pin inserter.
4. The sternal closure system as recited in claim 3, wherein the shaft portion of the pin inserter is accommodated within a cannulation of the molly anchor at a location between a first head portion of the molly anchor and a second head portion of the deployment pin.
5. The sternal closure system as recited in claim 4, wherein the deployment pin is preassembled to the molly anchor such that the second head portion of the deployment pin is countersunk within the cannulation relative to the first head portion.
6. The sternal closure system as recited in claim 4, wherein the second head portion of the deployment pin abuts against a circumferential ledge located within the cannulation of the molly anchor when the molly anchor is in the relaxed position.
7. The sternal closure system as recited in claim 6, wherein, when the locking tab is disengaged from the shaft of the pusher, the deployment pin is movable into engagement with the molly anchor to cause a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.
8. The sternal closure system as recited in claim 2, wherein the shaft extends from a proximal end section of the pusher, and further wherein the proximal end section includes first diameter that is greater than a second diameter of the shaft.
9. The sternal closure system as recited in claim 2, wherein the locking tab is configured to engage the shaft of the pusher to prevent the shaft from engaging the deployment pin.

- 10.** The sternal closure system as recited in claim 9, wherein, when the locking tab is disengaged from the shaft of the pusher, the deployment pin is movable into engagement with the molly anchor to cause a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.
- 11.** The sternal closure system as recited in claim 1, wherein the deployment pin is movably accommodated within a cannulation of the molly anchor.
- 12.** The sternal closure system as recited in claim 1, wherein the molly anchor includes a head portion and a body portion having a first deployable leg and a second deployable leg that each extend distally from the head portion.
- 13.** The sternal closure system as recited in claim 1, wherein the bone plate includes: a body including a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface; and a ridge protruding from the bone contacting surface and configured to align and center the bone plate relative to a bone.
- 14.** The sternal closure system as recited in claim 1, wherein the bone plate includes: a body having a first longitudinal section, a second longitudinal section, a bridging section extending between the first longitudinal section and the second longitudinal section, a bone contacting surface, and an outer surface opposite of the bone contacting surface; a first longitudinal slot provided within the first longitudinal section; and a second longitudinal slot provided within the second longitudinal section.
- 15.** The sternal closure system as recited in claim 1, comprising a holding device including a holding fork or a holding ball configured to engage the bone plate for holding the bone plate down against a bone.
- 16.** A surgical method, comprising: positioning a bone plate relative to a sternum; advancing a molly anchor of a fixation device deployment assembly through an opening of the bone plate and into the sternum while a locking tab is secured to the fixation device deployment assembly; removing the locking tab from the fixation device deployment assembly; and advancing a deployment pin to cause the molly anchor to transition from a relaxed position to a deployed position relative to the sternum.
- 17.** The surgical method as recited in claim 16, wherein advancing the molly anchor includes: applying a first force to a pusher of the fixation device deployment assembly with the locking tab secured to a shaft of the pusher.
- 18.** The surgical method as recited in claim 17, wherein advancing the deployment pin includes: applying a second force to the pusher with the locking tab removed from the shaft of the pusher.
- 19.** The surgical method as recited in claim 18, wherein applying the second force advances the deployment pin distally into a cannulation of the molly anchor to release the molly anchor from the fixation device deployment assembly.
- 20.** The surgical method as recited in claim 19, wherein advancement of the deployment pin distally into the cannulation causes a first deployable leg and a second deployable leg of the molly anchor to splay apart and thereby transition the molly anchor from the relaxed position to the deployed position.
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